$$\tilde{\mathbf{X}}^o = \mathbf{M}_B \odot \mathbf{X}_j^o + (1 - \mathbf{M}_B) \odot \mathbf{X}_i^o,$$

$$\tilde{\mathbf{X}}^s = \mathbf{M}_B \odot \mathbf{X}_j^s + (1 - \mathbf{M}_B) \odot \mathbf{X}_i^s,$$

$$\tilde{\mathbf{Y}} = \mathbf{M}_B \odot \mathbf{Y}_j + (1 - \mathbf{M}_B) \odot \mathbf{Y}_i.$$